

Intern Test Task: Configuration-Driven Data Collection and Analysis (Python)

Objective

Build a small, reproducible **Python project** that automatically collects **annual macroeconomic data** from the World Bank API using a **TOML configuration file**. The configuration must fully control what data is collected and over which time horizon. In addition to data collection and reshaping, you will perform a small descriptive analysis.

Expected Time Investment

Total: ~5-6 hours for a data-science student with some Python experience.

Repository Setup

Create a **public GitHub repository** containing:

- `README.md`
- `config.toml`
- Python source code
- One or more generated output files (data + analysis)

A fresh user should be able to:

1. Clone the repository
 2. Install dependencies
 3. Run a single command or script
 4. Obtain all outputs
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Configuration File (`config.toml`)

```
[countries]
list = [
    "Australia", "Belgium", "United States", "Canada", "Mexico", "Brazil",
    "Germany", "France", "Italy", "Denmark", "Finland", "Netherlands",
    "United Kingdom", "South Africa", "China", "Japan", "Turkey"
]

[time]
start_year = 2000
end_year = 2025

[series]
industry_value_added_usd_const = "NV.IND.TOTL.KD"
```

```
manufacturing_value_added_usd_const = "NV.IND.MANF.KD"  
gdp_usd_real = "NY.GDP.MKTP.KD"
```

Rules

- Only **annual data** may be used
- All changes to countries, years, or indicators must be done **only in the TOML file**
- Country names must be mapped to valid World Bank country codes

Data Collection Requirements

- Use Python (recommended libraries: `requests`, `pandas`, `tomllib` or `toml`)
- Correctly handle:
 - API pagination
 - Missing or partial observations
 - Countries or indicators with incomplete coverage
- Do not hard-code countries, indicators, or years in the source code

World Bank API documentation:

- <https://datahelpdesk.worldbank.org/knowledgebase/articles/889392>

Output Format

Produce one main dataset in **CSV** format with the following structure:

country	indicator	year_2000	year_2001	...	year_2025
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- `country`: country name or ISO code
- `indicator`: indicator key or ID
- Each year must appear as a **separate column**
- Missing values must be explicit (`NaN`)

Analysis Task

Using the collected data, perform a **descriptive analysis of structural change**.

Required Computations (per country)

1. Value-added shares

- Industry value added as a share of GDP
- Manufacturing value added as a share of GDP

2. Growth trends (2000–2025)

- Compound annual growth rate (CAGR) for:

- Industry value added
- Manufacturing value added
- GDP

3. Summary Table

Create a compact table with **one row per country**, including columns such as:

- `industry_share_2000`
- `industry_share_2025`
- `manufacturing_share_2000`
- `manufacturing_share_2025`
- `industry_cagr`
- `manufacturing_cagr`
- `gdp_cagr`

Export this table as CSV.

4. Short Interpretation

Include **5–10 sentences** in the README or a separate Markdown file answering questions such as:

- Which countries show signs of de-industrialization?
- Is manufacturing declining faster than total industry?
- Are trends different between advanced and emerging economies?

README Requirements

Your `README.md` should include:

- Project purpose
- Explanation of the TOML configuration
- Instructions to run the project
- Description of the output files
- Brief discussion of analytical findings
- Any assumptions or limitations

Evaluation Criteria

Data Engineering

- Correct and robust API usage
- Clean configuration-driven design
- Correct data reshaping

Data Analysis

- Correct computation of ratios and growth rates
- Sensible handling of missing data
- Clear descriptive interpretation

Software Practices

- Clear project structure
- Readable and commented code
- Reproducibility

Communication

- Clarity of the README
 - Explicit assumptions and design choices
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Deliverable

A GitHub repository link containing all required components.